

Video Article

Induction of Intestinal Graft-versus-host Disease and Its Mini-endoscopic Assessment in Live Mice

Vera Buchele^{1,3}, Maïke Büttner-Herold², Tina Vogler^{1,3}, Markus F. Neurath^{1,3}, Kai Hildner^{1,3}

¹Department of Internal Medicine 1, University Erlangen-Nürnberg, University Hospital Erlangen

²Institute of Pathology, Department of Nephropathology, University Hospital Erlangen

³Deutsches Zentrum Immuntherapie (DZI), University Hospital Erlangen

Correspondence to: Kai Hildner at Kai.Hildner@uk-erlangen.de

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Abstract

Acute graft-versus-host disease (GvHD) represents the most severe complication that patients previously undergoing allogeneic hematopoietic stem cell transplantation (allo-HCT) face and is frequently associated with a poor clinical outcome. While, for instance, GvHD manifestations of the skin are usually responsive to established immune-suppressive therapies and are, hence, not taking a fatal course, the presence and the intensity of intestinal GvHD, especially of the mid-to-lower parts of the gut, strongly influence the outcome and overall survival of patients with acute GvHD. Therapeutic options are essentially limited to the classic immune-suppressive agents yielding only moderate disease-mitigating effects. Hence, detailed knowledge about the tissue-resident immune cascade, changes in the intestinal microbiota, and the stromal response prior, upon, and after intestinal GvHD onset are urgently needed to understand the events and mechanisms underlying its pathogenesis and to develop innovative therapeutic options. Murine models of GvHD are frequently employed to identify and functionally assess molecules and pathways putatively driving intestinal GvHD. However, means to specifically monitor and evaluate intestinal inflammation over time are essentially lacking since established scores to assess and grade acute GvHD are routinely comprised of various parameters which rather reflect systemic GvHD manifestations. The detailed evaluation of intestinal GvHD has been restricted to studies using euthanized mice, thereby essentially excluding longitudinal (i.e., kinetic) analyses of the colonic compartment under a given experimental condition (e.g., antibody-mediated blockade of a proinflammatory cytokine) in live mice (i.e., in vivo). The mini-endoscopic in situ assessment of the distal colon of allo-HCT-treated mice described here allows a) a detailed macroscopic evaluation of different aspects of intestinal inflammation and b) the option to collect tissue samples for downstream analyses at various time points over the course of the observation period. Overall, the mini-endoscopic approach provides a major advance in preclinical noninvasive monitoring and assessment of intestinal GvHD.

Video Link

The video component of this article can be found at <https://www.jove.com/video/58871/>

Introduction

Hematopoietic malignancies directly arising from the hematopoietic stem cell compartment and uncontrolled, rapidly progressing, and severe immune-mediated disorders are often indications to perform allo-HCT^{1,2}. However, although accounting for the occurrence of the prognostic beneficial graft-versus-tumor response, donor lymphocytes are frequently inducing and promoting an unwanted immune-mediated attack of healthy tissue components within the allo-HCT recipient, a process that is called graft-versus-host disease³. Manifestations in the gut, the so-called intestinal GvHD, represent the most dreaded complication of acute GvHD, severe forms of which are routinely associated with a high mortality^{1,2,4}.

Overall, murine models of allo-HCT have emerged as invaluable tools to identify and study immune-mediated mechanisms underlying the pathogenesis of GvHD⁵. However, kinetic assessment of, for instance, beneficial effects of novel therapeutic interventions over time in live mice is routinely based on the determination of clinical GvHD scores⁶. While these scores are suitable to reflect, for instance, the overall disease burden (i.e., the systemic GvHD), clinical scores lack the sensitivity to reliably mirror organ-specific manifestations (e.g., in the gut). Hence, conclusions, for instance with respect to gut-protective effects of a given therapeutic intervention, that are based on these scoring systems usually fall short.

Despite major advances through the invention of novel whole-body imaging modalities in combination with the usage of either bioluminescent or fluorescent genetic mouse models^{7,8}, methodologies to directly and specifically assess the intestinal manifestation of GvHD in live mice are lacking. Hence, the rationale behind the protocol of the endoscopic assessment of the intestinal GvHD phenotype described in the next section is to overcome this obstacle. Furthermore, the motivation is also to reduce experimental mice numbers since, so far, a detailed assessment of

the cellular, morphological, and molecular characteristics (e.g., by histopathology or molecular biology) of intestinal GvHD manifestation has ultimately required the sacrifice of the experimental mouse.

Our institution has previously reported on the methodology of a mini-endoscopic assessment of colonic manifestations in the course of syngeneic colitis models⁹. In the protocol presented here, we have refined and adapted the colonoscopic scoring matrix for alloresponse-driven colitis in live mice with intestinal GvHD upon transplantation of alloreactive HCT and donor lymphocytes in an MHC class I fully mismatched setting. We identified four parameters suitable to reflect intestinal GvHD-related colonic lesions. Furthermore, we established a system that allows a fine-tuned grading of any single determinant, resulting in a new score that readily informs the reader about the severity of intestinal GvHD present in a given mouse at a given time point. Histopathological analyses confirmed that an endoscopic score above a certain threshold is reliably predicting moderate-to-high grade tissue inflammation. Hence, mini-endoscopic evaluation appears to represent a working substitute for the gold standard histopathology that routinely requires the sacrifice of the experimental mice. Importantly, this protocol can be applied at virtually any given time point and can be repeatedly used during the course of the disease^{10,11}. Furthermore, in contrast to the usage of bioluminescence-dependent approaches, no labor-intensive and time-consuming measures like intercrossing genetically modified mice are required and, hence, the methodology can be applied on virtually any mouse line of interest.

Taken together, given the detrimental clinical perspective of allo-HCT patients with severe intestinal GvHD, rapid scientific progress and more insight into the molecular mechanisms underlying the immune pathogenesis are urgently needed. Similarly, important, ethical considerations demand that the gain of knowledge should be achieved with the usage of the minimal number of experimental mice. Hence, both recognized claims on the research community exploring intestinal GvHD can be advanced by implementing serial mini-endoscopic evaluations of the colon in the experimental work chain to monitor and grade intestinal GvHD in live experimental mouse models, as described and validated in the protocol presented here.

Protocol

The experimental methods described here have been approved by the government of Mittelfranken, Bavaria, Germany.

1. GvHD induction

1. Day 0: Total body irradiation of the recipient mice

1. Use female CD45.2⁺ H2kd⁺ BALB/c mice that are at least 10 weeks old as recipients.
2. Weigh and record the weight of the recipient mice prior to GvHD induction.
NOTE: Ensure that the recipient has a minimal body weight of 20 g. The starting body weight will serve as the reference value to calculate the weight loss of the individual mouse over the course of GvHD induction and progression.
3. Place up to five mice in the container of the X-ray irradiator.
4. Irradiate the mice by total body irradiation with a single dose of 8 Gy of X-ray. Use Cs¹³⁷ as a radiation source.

2. Day 1: Reconstitution of the irradiated mice with T-cell-depleted bone marrow

NOTE: The transplantation of allogeneic bone marrow cells, as outlined and detailed under section 1.2, should take place within 24 h after irradiation. Perform the procedures described below in a sterile tissue culture hood and use filtered reagents. Use allogeneic CD45.1/Ly5.1 B6.SJL-*Ptpca Pepcb*/BoyCrl mice as donors for T-cell-depleted bone marrow.

1. Euthanize CD45.1/Ly5.1 B6.SJL mice that will donate allogeneic bone marrow cells in accordance with the institutional guidelines, 12–24 h after the irradiation of the recipient BALB/c mice. For this, anesthetize the CD45.1/Ly5.1 B6.SJL mice by inhalation of 5% isoflurane. Continue with the isoflurane exposure over 1 min after breathing arrest and confirm euthanasia by cervical dislocation.
2. Disinfect the fur and skin of the mouse thoroughly with 70% ethanol.
3. Position the mouse onto a clean working sheath so that the mouse is in a prone position, with its hindquarters facing the experimenter. Lift the fur at the Achilles' heel with the tip of a Semken forceps with a length of 13 cm and serrated curved tips. Incise the skin between the forceps and the heel with hardened 8.5 cm fine scissors with straight tips.
4. Elongate the incision cranially (i.e., from the heel over the lower leg and thigh to the hip region). Remove the skin and fur from the hind limb with the help of the forceps.
5. Perform the same procedure on the other hind limb.
6. Remove the hind limbs by cutting through the adjacent hip joints.
7. Cut away the rear paws. Cut through the knee joint. Store the thigh and shank of every hind limb together in a 92 mm Petri dish filled with phosphate-buffered saline (PBS) solution on ice.
8. Clean the femur and tibia bones carefully by removing as much muscle tissue as possible from every thigh and shank. Transfer the femur and tibia bones to a 50 mL tube filled with sterile RPMI 1640 medium and place it on wet ice until all tibia and femur bones collected in step 1.2.7 are cleaned from muscle tissue.
9. To isolate the bone marrow, transfer one femur or tibia bone at a time (which has been cleaned as described in step 1.2.8) from the 50 mL tube to a Petri dish (with a diameter of 92 mm) filled with RPMI 1640 medium. Cut off the ends of the femur or tibia with a scalpel to get access to the cavity of the bone containing the bone marrow.
10. Prepare a 50 mL collection tube with 5 mL of RPMI 1640 medium. Insert a 26 G needle attached to a 1 mL syringe filled with 1 mL of RPMI 1640 medium into the bone cavity and flush the bone marrow out of the cavity into the collection tube by pushing the plunger.
11. Repeat step 1.2.10 until the bone appears light and shiny (i.e., the bone cavity is visibly devoid of reddish bone marrow). Discard the empty bone.
12. Repeat steps 1.2.9 – 1.2.11, using all femur and tibia bones from step 1.2.8, and collect all recovered bone marrow pieces in the same 50 mL collection tube from step 1.2.10. Keep the collection tube on wet ice until all bones from step 1.2.8 have been processed accordingly.
13. Create a single-cell suspension by softly pipetting the flushed, crumbly bone marrow pieces up and down. Filter the bone marrow single-cell suspension through a 40 µm mesh screen cell strainer placed on a new 50 mL collection tube.

14. Centrifuge the 50 mL collection tube containing the bone marrow single-cell suspension at 450 x g for 5 min at 4 °C. Discard the supernatant.
15. Resuspend the pellet in 5 mL of ACK lysing buffer (1 mM Na₂EDTA; 10 mM KHCO₃; 144 mM NH₄Cl; pH 7.2) for red blood cell lysis. Incubate the cell suspension for 3 min at room temperature. Afterward, immediately add 10 mL of PBS solution to the cell suspension.
16. Centrifuge the suspension at 450 x g for 5 min at 4 °C. Discard the supernatant and resuspend the cells in 2 mL of PBS solution.
17. Count the bone marrow cells, using a hemocytometer. Preserve an aliquot of 6 x 10⁶ cells for flow cytometry analysis to check the purity of the cells after cell purification (see step 1.2.20).
18. For the depletion of T cells from the bone marrow single-cell suspension, use a commercially available cell purification kit. Magnetically deplete CD90.2⁺ cells (i.e., lymphocytes) from the total bone marrow cells, following the manufacturer's protocol.
19. Count the T-cell-depleted bone marrow cells that have been isolated as described in step 1.2.18, using a hemocytometer. Preserve an aliquot of 1 x 10⁶ cells for flow cytometry analysis to be performed later on, as described in step 1.2.20.
- NOTE:** The on-average cell yield derived from one donor mouse is usually from 2 x 10⁷ to 3.4 x 10⁷ T-cell-depleted bone marrow cells.
20. Confirm a successful T-cell depletion by flow cytometry. Stain 1 x 10⁶ cells, preserved from steps 1.2.17 (bone marrow cells before magnetic cell separation) and 1.2.19 (T-cell-depleted bone marrow cells after magnetic cell separation), with the following antibodies: α-CD45.1 (A20), α-CD3 (17A2), α-CD4 (GK1.5), and α-CD8α (53-6.7). Use the remaining cells from step 1.2.17 as an unstained control and for single-staining controls, to set up the flow cytometer.
 1. Transfer 1 x 10⁶ cells from each sample to a well of a 96-well plate with a V-bottom to perform the flow cytometric staining procedure.
 2. Spin down the cells within the 96-well plate (step 1.2.20.1) at 450 x g for 5 min at 4 °C. Discard the supernatant and resuspend the cells in 100 μL of FACS buffer (PBS supplemented with 3% filtered bovine serum).
 3. Centrifuge the cells within the 96-well plate at 450 x g for 5 min at 4 °C. Discard the supernatant.
 4. Add an appropriate amount of the antibody of choice (compare with step 1.2.20) to the samples for single staining in 100 μL of FACS buffer.
 5. Prepare a mixture of all antibodies (master mix) containing the appropriate amounts sufficient to separately stain the bone marrow cell aliquots preserved from prior (see step 1.2.17) and after (see step 1.2.19) magnetic T-cell depletion. Add 100 μL of the master mix to both aliquots.
 6. Add only 100 μL of FACS buffer to the unstained sample.
 7. Incubate the cells within the 96-well plate for 20 min at 4 °C in the dark.
 8. Add 100 μL of FACS buffer and centrifuge the cells within the 96-well plate at 450 x g for 5 min at 4 °C. Discard the supernatant and resuspend the cells in 100 μL of FACS buffer.
 9. Centrifuge the cells at 450 x g for 5 min at 4 °C. Discard the supernatant and resuspend the cells in 250 μL of FACS buffer.
 10. Transfer the samples to a 5 mL polystyrene round-bottom tube and analyze them with a flow cytometer instrument that is able to detect the fluorescent molecules that were used to label the antibodies employed to characterize the cell samples (see step 1.2.20).
 11. Compare the cell composition from before and after the magnetic cell separation.
- NOTE:** A purity of approximately 95% CD45.1⁺CD3⁺ (i.e., T-cell-depleted bone marrow) cells within the live gate is usually achieved.
21. Wash the T-cell-depleted bone marrow cell suspensions 2x with PBS solution (450 x g for 5 min at 4 °C) and, finally, resuspend the cells in PBS solution, adjusting the cell concentration to 5 x 10⁷ cells/mL. Keep the cells on wet ice until the injection.
22. Inject T-cell-depleted bone marrow cells into the recipient mice, which were irradiated the day before (see section 1.1).
 1. Place the experimental mouse in a leak-proof chamber and induce anesthesia by inhalation of up to 4% isoflurane, until the mouse is unconscious. Confirm analgesia and loss of consciousness by testing the loss of the righting reflex and the subsequent loss of the pedal withdraw reflex.
 2. Inject 5 x 10⁶ T-cell-depleted bone marrow cells (i.e., 100 μL of 5x10⁷ cells/mL) containing PBS solution using a 1mL syringe equipped with a 30 G needle intravenously into the retrobulbar space containing the venous sinus.

3. Day 2: Transfer of T cells

NOTE: Perform the described procedures below in a sterile tissue culture hood and use filtered reagents. Use CD45.2 C57Bl/6 (wild-type; WT) mice as donors for alloreactive T cells. The usage of congenic marker systems allows the distinguishability between recipient (CD45.2⁺ H2kd⁺ Balb/c mice) and donor hematopoietic cell types (bone marrow cells: CD45.1⁺ H2kb⁺ B6.SJL mice; allogeneic T cells: CD45.2⁺ H2kb⁺ C57Bl/6 mice).

1. Euthanize C57Bl/6 mice in accordance with the applying institutional guidelines and authorities. Therefore, anesthetize mice by inhalation with a concentration of 5% isoflurane. Continue the isoflurane exposure until 1 min after breathing arrest and confirm euthanasia by cervical dislocation. Disinfect the fur and skin of the mouse thoroughly with 70% ethanol.
2. Place a strainer with a 40 μm mesh screen on a 50 mL collection tube. Remove the spleen and place it onto the strainer.
3. With a syringe plunger, comminute the spleen in the strainer. Wash the strainer and syringe plunger with PBS to collect all splenocytes.
4. Centrifuge the cells within the 50 mL collection tube at 450 x g for 5 min at 4 °C. Discard the supernatant.
5. Resuspend the splenocytes in 3 mL of ACK lysing buffer to lyse red blood cells. Incubate the cell suspension for 3 min. Afterward, add 10 mL of PBS and centrifuge the cells at 450 x g for 5 min at 4 °C. Discard the supernatant and resuspend the splenocytes in PBS.
6. Count the spleen cells, using a hemocytometer. Divert an aliquot of 6 x 10⁶ cells for flow cytometry analysis, to evaluate the magnitude of purification in step 1.3.9.
7. For a total CD3⁺ T-cell isolation from total splenocytes, use a commercially available cell purification kit. Isolate the splenic T cells, following the manufacturer's protocol.
8. Count the splenic CD3⁺ T cells isolated as described in step 1.3.7, using a hemocytometer. Preserve an aliquot of 1 x 10⁶ T cells for flow cytometry analysis.

NOTE: The on-average yield of splenic T cells per donor mouse isolated by magnetic cell separation ranges from 1.3 x 10⁷ to 2 x 10⁷ cells.

9. Confirm the success of the T-cell isolation by flow cytometry. For a detailed description of the staining protocol, confer and follow steps 1.2.20.1 - 1.2.20.10.
 1. Stain 1×10^6 splenocytes of steps 1.3.6 (before the magnetic cell separation) and 1.3.8 (after magnetic cell separation) with the following antibodies: α -CD45.2 (104), α -CD3 (17A2), α -CD4 (GK1.5), and α -CD8 α (53-6.7).
 2. Use the remaining cells from step 1.3.6 as an unstained control and for single staining with the respective antibodies, to set up the flow cytometer.
 3. Compare the cell composition before and after magnetic cell separation.
NOTE: A purity of $\geq 95\%$ of CD45.2⁺CD3⁺ T cells within the lymphocyte live gate should be accomplished.
10. Wash the T cells 2x with PBS (450 x g for 5 min at 4 °C) and, finally, resuspend the cells in PBS solution, adjusting the cell concentration to 7×10^6 cells/mL. Keep the cells on ice until the injection.
11. Inject alloreactive, splenic T cells (see step 1.3.10) into irradiated BALB/c mice who have previously received CD45.1⁺ T-cell-depleted bone marrow cells (see step 1.2.22).
 1. Induce anesthesia by placing the experimental mouse in a leak-proof chamber in which it is exposed to up to 4% isoflurane until it loses its consciousness. Confirm analgesia and loss of consciousness by testing the loss of the righting reflex and the subsequent loss of the pedal withdraw reflex.
 2. Inject 0.7×10^6 alloreactive CD3⁺ T cells using a 1 mL syringe equipped with a 30 G needle (*i.e.*, 100 μ L of 7×10^6 cells/mL) containing PBS solution (see step 1.3.10), intravenously into the retrobulbar space of the mice, to induce GvHD. Denominate these mice as the experimental group.
 3. Inject a different group of mice with 100 μ L of PBS alone, intravenously into the retrobulbar space, and denominate this group as the control group, subsequently called "no-T-cell-transplanted (noT) mice".

2. Assessment of Systemic GvHD

NOTE: Perform this procedure in a semi-sterile surrounding in an experimental room licensed and equipped for handling live mice within the animal facility.

1. **Follow the clinical course of GvHD in individual mice; specifically, assess and score the experimental mice 3x a week for the presence of clinical GvHD symptoms, based on a scoring system described below and adapted from a scoring system established previously by Cook et al.⁶.**
 1. Weigh each mouse individually, record the weight, and calculate the specific body weight loss in reference to the body weight that was determined prior to the start of the experiment (corresponding to 100%) on day 0 (step 1.1.2). Score a weight loss of less than 10% of the starting weight with grade 0, a weight loss between 10% and 25% with grade 1, and a weight loss of more than 25% with grade 3.
 2. Score the activity level of each mouse, discriminating mice with a normal activity level (score = 0) from mice with a moderately decreased activity level (score = 1) and mice with stationary behavior unless they are externally stimulated (score = 2).
 3. Score the fur texture, thereby discriminating a normal, shiny, and kempt fur (score = 0) from a moderately ruffled fur (score = 1) and the presence of bald spots (score = 2).
 4. Score the skin texture, discriminating normal skin (score = 0) from sporadic spots of scaling skin (*e.g.*, at the tail and the paws) (score = 1) and obvious scaling and sore skin areas (score = 2).
 5. Analyze the consistency of secreted stool fractions. Score stool with a normal (*i.e.*, hard consistency) with grade 0, stool with deformable and soft stool with grade 1, and stool without any consistency and, hence, with severe diarrhea with grade 2.
 6. Sum up all separately scored parameters to a maximal clinical score of 12 per mouse.
2. Consider and evaluate the criteria of ceasing the experiment due to the current disease severity level of the individual mouse, in accordance with the applying institutional guidelines and authorized animal protocol.

3. Assessment of Intestinal GvHD

1. **Scoring of GvHD-related lesions by mini-endoscopy of the distal colorectal region**
NOTE: Perform this in a semisterile surrounding in an experimental room licensed and equipped for handling live mice within the animal facility. Use a mini-endoscopic workstation that is approved for the use of small animals.
 1. Prepare the endoscopic device by covering the telescope (with a diameter of 1.9 mm and a length of 10 cm) with the endoscopic examination sheath. Clean and sterilize the endoscope with water and ethanol.
 2. Switch on the computer, the adjustable xenon light source, and the camera module, and set the focus in a way that objects close to the lens give a clear image.
 3. Connect the air pump to the endoscopic sheath. To visualize the air stream, dip the tip of the endoscope into a beaker glass filled with water. Regulate the air stream by adjusting the valve between the air pump and the endoscopic sheath. Adjust it to a slow, constant air stream, reflected by a few continuously ascending bubbles.
 4. Place a mouse into a leak-proof chamber. Induce anesthesia by inhalation of up to 4% isoflurane until the mouse is unconscious. Confirm analgesia and an absent state of the reflexes by testing the loss of the righting reflex and the subsequent loss of the pedal withdraw reflex.
 5. Use an appropriate veterinary anesthesia machine in accordance with the animal protocol used. Transfer the mouse from the chamber to the colonoscopy workspace. Here, position the anesthetized mouse, with its nose in the nosecone (which is connected to the anesthesia instrument), onto a clean working sheath so that the mouse is in a prone position, facing the experimenter with its hindquarters.

6. To maintain anesthesia during the colonoscopy procedure, reduce the isoflurane concentration to approximately 2%. Monitor the mouse's respiration and response to stimulation during the colonoscopy and adjust the isoflurane concentration at the vaporizer if needed. Cover the mouse's eyes with saline to prevent them from getting dry.
7. Control the efficacy of the anesthesia by pinching between the toes of the mouse. In the case of absent reactivity, proceed to step 3.1.8.
8. Lift the tail just above the tail root with one hand and carefully insert the endoscope, positioned in the other hand, via the anus into the rectum.
9. While the air stream inflates the colorectal lumen, slowly and carefully move the endoscope forward in aboral direction.
10. Place the endoscope in the middle of the colonic lumen and maintain this position in order to minimize contact with the colonic wall and to avoid making scratches, deeper wall-related injuries (e.g., resulting in bleeding), or even wall perforation (see step 3.1.17). Control this step by permanently watching the video stream (i.e., under constant visualization of the colonic compartment).
11. If feces constrict the view, remove the endoscope and perform an enema by rectally flushing with up to 2 mL of saline, using a soft Pasteur pipette. Use as little liquid as possible, to prevent an unintended dilution of the feces and other secondary effects negatively affecting the characteristics of the mucosal surface and, ultimately, the scoring results.
12. Try to regularly position the endoscope at a defined (i.e., distinct) anatomic site within the distal colon to standardize the scoring of colonic inflammation and to optimize the comparability of scoring results between mice and across experiments.
NOTE: Normally, with the rigid endoscope, a maximal depth of 4 cm can be reached via the rectal route. Between 2 and 4 cm aborally, the colon regularly turns into a flexure that cannot be passed with the rigid colonoscope. Use the colon region distally adjacent to the flexure as the default scoring spot.
13. Start to record the video stream and begin with the scoring of inflammation.
14. Evaluate the GvHD-related inflammation of the colon by scoring the parameters explained and depicted in **Table 1** and **Figure 3**.
 1. Move the endoscope at the scoring spot gently and slightly back- and forward to assess the different parameters.
 2. Assess the parameter "translucency", as well as the "stool consistency", by positioning the endoscope in a wider distance in relation to the colonic wall.
 3. Assess the parameters "granularity" and "vascularity" by positioning the endoscope in close proximity to the colonic wall. For this task, carefully apply well-dosed tension to the colonic wall with the tip of the endoscope.
15. Upon completion of the scoring process, stop recording.
NOTE: Afterward, the recorded video clip can be used in total or used to generate representative images (i.e., screenshots) of the mini-endoscopically assessed criteria overall contributing to colonic inflammation.
16. Carefully remove the endoscope.
17. Discontinue isoflurane exposition by transferring the mouse to a separate cage in which it is exposed to ambient air. Warm the mouse with a red-light lamp and observe the mouse until it recovers from the anesthetic and regains full consciousness.
NOTE: Due to the air inflation of the colon during the endoscopy, the abdominal region of the mouse may appear puffed up directly afterward. While this is normal and of transient nature, prolonged signs of massive and even progressive inflation of the abdomen and failure of recovery of the mouse might indicate an unintended perforation of the colon (in this case, the mouse needs to be euthanized immediately).
18. Upon complete recovery, place the mouse back into its respective cage.
19. Optional approach: It is also possible to take view-guided tissue biopsies. Take the following steps.
NOTE: The colonoscopy will be performed in the same way as described in steps 3.1.1 - 3.1.18, with the following modifications.
 1. In step 3.1.1, use an endoscopic examination sheath with a built-in working channel, instead of a simple examination sheath without working channels, for covering the telescope. Introduce a biopsy forceps into the working channel until its tip is just visible in front of the endoscope at the video screen because this largely prevents unintended injury to the intestine.
 2. Proceed with steps 3.1.2 - 3.1.11. Insert the endoscope into the colon and push it carefully forward to the spot of interest.
 3. Employ the help of a second experimenter for the task of taking colonic tissue biopsies. Ask the second experimenter to navigate the tip of the biopsy forceps to the colonic region of choice by pushing the forceps within the working channel slowly forward until the jaws of the forceps can be opened. Constantly watch the video stream during this procedure to minimize the risk of injuring the colonic wall.
 4. Recover a colonic wall tissue sample by carefully opening and closing the jaws of the biopsy forceps.
NOTE: Do this with caution to prevent the perforation of the colonic wall. In the rare case of a colonic perforation, immediately sacrifice the affected mouse by applying measurements in accordance with the institutional guidelines and under the continuous administration of anesthetics.
 5. Pull back and remove the closed forceps out of the working channel. Remove the specimen from the jaws of the biopsy forceps by dipping and carefully shaking the opened forceps in sterile PBS solution. Afterward, choose proper storage conditions for the tissue sample, in accordance with the planned downstream analyses. After disinfection with 70% ethanol, the forceps might be reused to take further biopsies.
 6. After taking biopsies, remove the endoscope and finish the colonoscopy as described in steps 3.1.16 - 3.1.18.

2. Histopathological analysis

1. Between days 26 and 30 after the X-ray irradiation, euthanize the allo-HCT recipient mice in accordance with applying institutional guidelines and authorities. For this, anesthetize the mice by inhalation of 5% isoflurane. Continue with the isoflurane exposure for 1 min after breathing arrest and confirm euthanasia by cervical dislocation. Thoroughly disinfect the fur and skin of the mouse with 70% ethanol.
2. Open the abdominal cavity and remove the colon. Transfer it to a Petri dish (with a diameter of 92 mm) with PBS.
3. Fill a 5 mL dispenser tip with PBS. With the help of a Semken forceps with a length of 13 cm and serrated curved tips, slip the distal part of the colon (approximately 5 mm) over the tip of the dispenser tip. Carefully fix the colon with the forceps at the dispenser tip and flush the colon with PBS by pressing down the plunger of the dispenser tip to remove feces from the intestine.
4. Cut out a colonic segment located about 5 mm from the anus, with the help of a scalpel.
5. Fix the resected gut specimen in 4.5% formaldehyde overnight. Before embedding it in paraffin, place the tissue in an upright position.

6. Cut 3 μm cross sections of paraffin-embedded colon tissue and stain them with hematoxylin and eosin (HE), using standard staining protocols.
7. Prepare HE-stained cross sections of colon tissue samples.
8. Adapted from the scoring matrix reported by Kaplan *et al.*¹², histopathologically score the inflammatory activity semiquantitatively, with scores from 0-3: no inflammation (score = 0), mild inflammation (score = 1), moderate inflammation (score = 2), and severe inflammation (score = 3).

NOTE: Ideally, employ for this task the expertise of a pathologist who is experienced in the evaluation of murine intestinal GvHD and blinded to the nature of the experimental and control groups.

Representative Results

The current protocol, describing the mini-endoscopic evaluation of intestinal GvHD-associated lesions of the distal colon, has been established and validated in mice previously subjected to the systemic induction of a severe acute GvHD model. In this study, we used an MHC class I fully mismatched model system in which BALB/c mice were lethally irradiated, followed by the transplantation of T-cell-depleted allogeneic bone marrow and by the administration of GvHD-inducing alloreactive C57BL/6 CD3⁺ T lymphocytes. T-cell-depleted bone marrow cells and splenic T cells for GvHD induction were purified by magnetic separation. The purity of these cell populations was determined by flow cytometry, and representative results of the purification process are displayed in **Figure 1**, showing a sufficient depletion of T cells from total bone marrow cells and a consistent enrichment of spleen-derived CD3⁺ T cells prior to their transfer to previously irradiated mice. The clinical course displayed in **Figure 2** demonstrates the reproducibly robust induction of the clinical systemic GvHD phenotype upon allo-HCT in the presence (WT, in black) vs. absence (noT, grey) of alloreactive donor T lymphocytes. The scoring system previously reported by Cooke *et al.* represents a sum core in which six parameters are assessed and graded: body weight, posture, activity, skin and fur texture, and stool consistency⁶. The control mice experienced only one, early occurring peak of the clinical score, between days 5 and 10, that was similarly observed in T-cell-receiving mice and is, therefore, largely due to the irradiation-associated systemic inflammatory response. Regardless, clinical scores of donor T-cell-receiving mice started to rise earlier, showed a higher total peak, only transiently decreased after initially peaking, and then, essentially increased again, continuously, over the remaining period of the experiment. Overall, these results are in agreement with the interpretation that T-cell-receiving mice show stronger and more progressive signs of systemic GvHD compared to noT mice.

The donor T-lymphocytes-receiving allo-HCT mice showed various signs of acute organ-related and systemic GvHD manifestations, overall resulting in high sum scores, while the control mice lacked moderate to severe affections, especially at later time points. However, signs of intestinal GvHD are underrepresented in the systemic GvHD scoring system, where the only assessed gut-related parameter is stool consistency. As shown in **Table 1**, mini-endoscopically assessable criteria were defined, specifically intended for the precise description, scoring, and grading of intestinal GvHD-associated lesions, by adapting criteria previously reported for the evaluation of syngeneic colitis to the context of alloresponse-driven colitis⁹. **Figure 3** displays typical examples for each individual criterion, illustrating the type and extent of intestinal GvHD-associated lesions, thereby visualizing the grading matrix applied during the mini-endoscopic evaluation of the distal colon of GvHD-prone mice upon transplantation of MHC class I fully mismatched donor lymphocytes.

Figure 4A shows that intestinal GvHD sum score results that are based on criteria defined in **Table 1** and displayed in **Figure 2** easily enable the experimenter to discriminate donor lymphocyte-receiving mice with severe signs of intestinal inflammation from control mice that are essentially devoid of GvHD. To validate the mini-endoscopically based grading system, histopathological studies were performed by using a previously reported microscopic grading system¹². The data in **Figure 4B** confirm that the colon of mice severely affected by GvHD-related inflammation, as evidenced by high colonoscopic sum scores, display similarly strong histopathological signs of inflammation, given a histopathological sum score of ≥ 2 postmortem. In contrast, colon tissues of control mice display no (score 0) or, at most, mild (score 1) histopathological signs of inflammation, in line with the virtual absence of mini-endoscopically detectable signs of colitis. Furthermore, as shown in **Figure 4C, D**, correlation studies between mini-endoscopically and histopathologically assessed colitis activity and systemic GvHD scores were performed. Importantly, these studies demonstrated that mini-endoscopically determined sum scores of ≤ 3 reliably predict the absence of mid-to-higher grade (i.e., ≥ 2) intestinal GvHD-associated colonic inflammation scores obtained from histopathological grading. Finally, the severity of endoscopically assessed intestinal GvHD shows a correlation with the systemic GvHD activity.

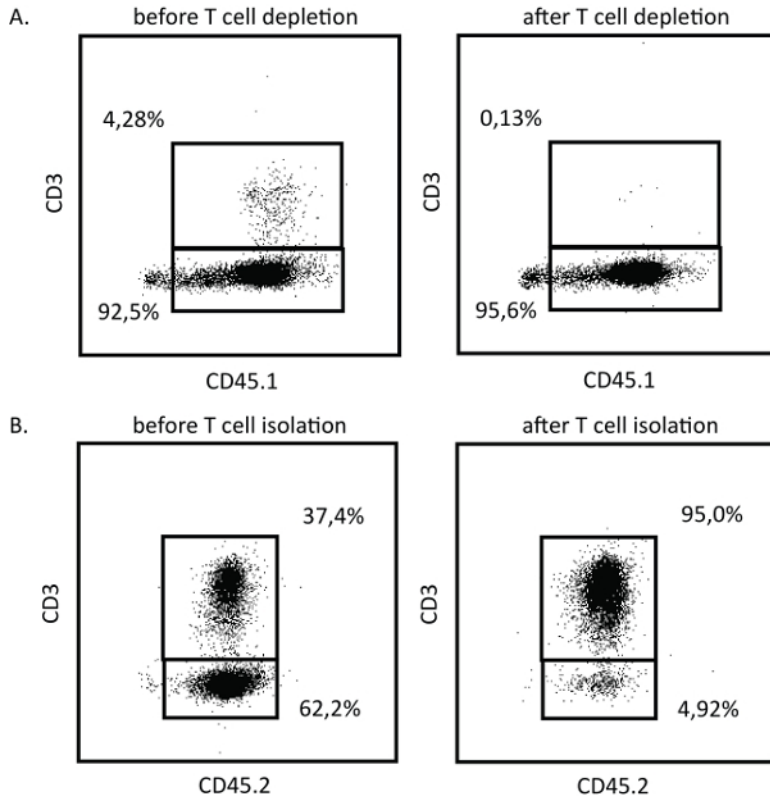


Figure 1: Flow cytometric quality assessment of magnetically purified T-cell-depleted bone marrow cells and allogeneic splenic CD3⁺ T cells. (A) Depletion of CD90.2⁺ bone marrow cells achieved by magnetic separation, using a commercially available purification kit. The purity of T-cell-depleted bone marrow cells was determined by flow cytometry, by staining cells prior to and after T cell depletion with anti-CD45.1 and anti-CD3. Results from one representative experiment are shown. (B) Splenic T cells were magnetically purified, employing a commercially available purification kit. The purity was determined by flow cytometry, comparing the frequencies of CD45.2 and CD3 costaining of samples derived from before and after T cell isolation. Results from one representative experiment are shown. [Please click here to view a larger version of this figure.](#)

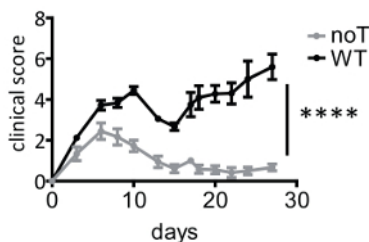


Figure 2: Clinical course of GvHD, employing an MHC class I fully mismatched model. To induce acute GvHD, BALB/c mice were whole-body irradiated on day 0. The mice received T-cell-depleted bone marrow cells 24 h later (d1). On day 2, the mice were injected with allogeneic splenic CD3⁺ T cells (wild-type [WT]; $n = 9$). As a control, some mice received T-cell-depleted bone marrow alone (noT; $n = 9$). The mice were specifically assessed three times a week for the presence and severity of clinical symptoms of GvHD. The displayed data represent mean values $\pm$ SEM of the clinical scores obtained from individual mice of the indicated treatment group from two representative experiments. The data were analyzed by two-way ANOVA, followed by Bonferroni's multiple comparisons posttest. **** $p < 0.0001$ was considered significant. [Please click here to view a larger version of this figure.](#)

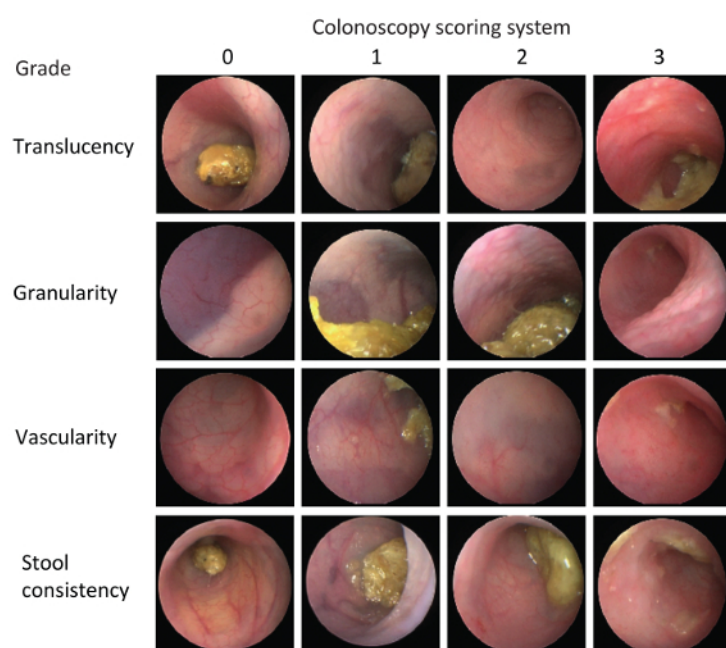


Figure 3: Representative images illustrating grading examples underlying the scoring matrix of the intestinal GvHD-related colon alterations assessed by a mini-endoscopic evaluation of the colons of live, allo-HCT-treated mice. To complement and illustrate the detailed description of the used scoring grading and system in **Table 1**, representative mini-endoscopic images of the evaluated severity levels (grading) for all parameters individually in the colon of anesthetized, live mice undergoing allo-HCT between days 26 and 30 before as described in **Figure 1** are displayed here. [Please click here to view a larger version of this figure.](#)

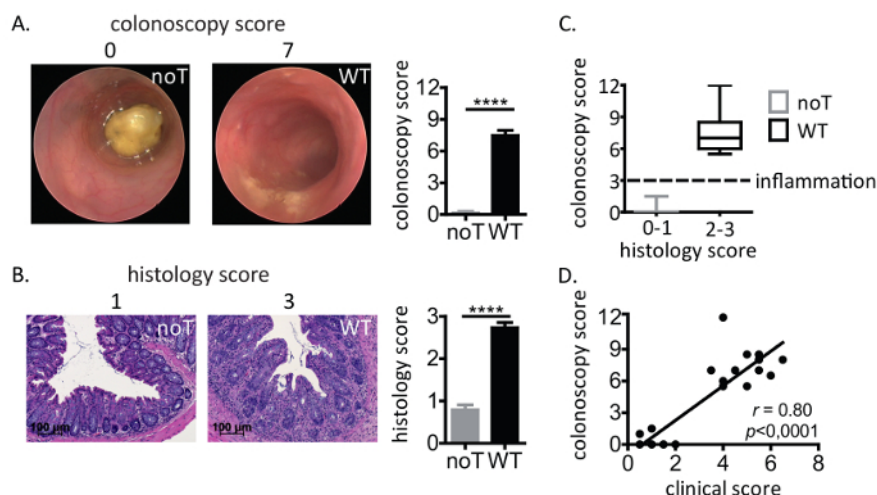


Figure 4: Mini-endoscopic scoring and grading of intestinal GvHD-related inflammation of the colon, reliably predicting the presence of higher-grade colitis assessed by histopathological scoring. (A) Between days 26 and 29 after the induction of GvHD as described in Figure 1, the manifestation of intestinal GvHD was evaluated by mini-endoscopy in live, anesthetized mice. Colonic alterations were scored and graded according to the scoring and grading system depicted in Table 1 and Figure 2. Representative endoscopic images of control (no T cell; noT) and wild-type (WT), T-cell-receiving mice and the mini-endoscopically assessed score (top) are shown. (B) The mice were sacrificed 12 h after the mini-endoscopic evaluation, and the distal part of the colon was processed and histopathologically assessed. The inflammatory activity of hematoxylin- and eosin-stained cross sections of the distal colon was graded. Representative hematoxylin- and eosin-stained cross sections of the distal colon of no T-cell- and WT T-cell-receiving allo-HCT-treated mice and the corresponding histology scores are shown. The data in panels A and B were analyzed by Student's *t*-test and are shown as mean $\pm$ SEM. *****p* < 0.0001 was considered significant. WT: *n* = 15; noT: *n* = 15. The data represent pooled data from four individual experiments. (C) Interdependence of colonoscopy scoring and histological scoring. The data are shown as box plots. Each blot displays the median (black line within each box), the range of the data (min to max), and quartiles (areas of each box). WT: *n* = 15; noT: *n* = 15. The data represent pooled data from four individual experiments. (D) Correlation between colonoscopy scoring and clinical scoring. Both control and wild-type T-cell-receiving mice are included in the correlation analysis. The solid line represents the linear regression line. Spearman correlation coefficient (*r*-value) and *p*-value are shown. WT: *n* = 12; noT: *n* = 13. The data represent pooled data from three individual experiments. [Please click here to view a larger version of this figure.](#)

Score	0	1	2	3
Translucency of the colonic wall	Extra-intestinal, inner organs (e.g. spleen) thoroughly visible	Discrete reduction of the visibility of extra-intestinal organs due to mild opacity of the colonic wall	Moderate reduction of the visibility of extra-intestinal organs due to significant opacity of the colonic wall	Lack of the visibility of extra-intestinal organs, i.e. non-transparent colonic wall
Granularity	Smooth, unaffected mucosal surface appearance; colonic crypt pattern visible	Discrete roughening and cobblestone appearance of the mucosal surface	Moderate roughening and cobblestone appearance of the mucosal surface	Severe roughening and cobblestone appearance of the mucosal surface; cushion-like appearance of the mucosa
Vascularity	Unaltered vascular pattern displaying the communicating network of large and small vessels	Discrete alterations of the vascular pattern; vessel pattern seems to fray	Some vessels are invisible; discontinuous vessel network	Contact induced bleeding; dot-like pattern of the vessels
Stool consistency	Normal; stool is hard; fine threads of mucus between feces and colonic wall can be observed	Stool is still shaped but may contain ragged edges; deformable with the tip of the endoscope	Stool is soft and unshaped; stool is visibly more shining (higher water content)	Stool is loose, liquid; watery diarrhea. Stool may be randomly spread over the mucosal surface of the colon

Table 1: Mini-endoscopic scoring and grading matrix of intestinal GvHD-related lesions in the colon of live allo-HCT-treated mice.

Alterations of the endoluminal and transmural colon morphology in allo-HCT pretreated mice were assessed by a colonoscopy of the rectum and the distal colon of anesthetized, live mice, using a murine mini-endoscopy system. Colonic lesions were classified with the help of the endoscopic scoring system (modified murine endoscopic index of colitis severity [MEICS]) that is deduced from the original MEICS scoring system reported by Becker et al.⁹ and adapted to the context of allo-HCT. The distal colon is visually assessed for the presence and magnitude of the following four parameters: 1. the transmittance of endoscopic light (translucency) through the colonic bowel wall as thickening of the wall; 2. the mucosal surface displaying a cobblestone appearance (granularity) of the endoluminal-oriented mucosal surface; 3. an altered vascularization pattern (vascularity); 4. endoscopically assessed stool appearance in situ (stool consistency). Each parameter is scored from 0 (no signs) to 3 (most severe phenotype), adding up to a maximum sum score of 12 per mouse.

Discussion

The protocol describes the methodology of induction and mini-endoscopic assessment of the colonic phenotype observed in the course of intestinal GvHD. It serves the wider purpose of enabling scientists to study intestinal GvHD longitudinally and noninvasively over the entire disease course (i.e., from the onset of colonic manifestation and progression until maximal disease activity).

However, there are some critical steps and important limitations inherent to the presented methodologies the experimenter needs to be aware of prior to applying the technologies in the respective scientific context. First, although we have successfully used the mini-endoscopic assessment of intestinal GvHD-related colonic lesions in a minor histocompatibility-mismatched model system¹¹, the provided protocol here is currently exclusively applicable to MHC class I fully mismatched acute GvHD models. While the MHC class I fully mismatched GvHD model is common and frequently used⁵, it is well established that the characteristics of GvHD manifestation are highly dependent on the used mouse substrains since this strongly impacts, for instance, the sensitivity to irradiation. Furthermore, the experimental mouse source with, for instance, a varying composition of intestinal microbiota and used numbers of transferred allogeneic T cells may critically impact the kinetics and dynamics of the intestinal GvHD manifestation¹². Hence, a critical step is to carefully test and validate the indicated measures for their ability to successfully induce intestinal GvHD as shown.

Regarding the successful implementation of the live mini-endoscopic assessment of acute intestinal GvHD, we want to stress that, while handling the endoscope is straightforward, it may require some training, to minimize the risk of experiencing the most dreaded complication (i.e., to perforate the intestinal wall with the rigid instrument in combination with the need to air-inflate the colon). Due to an increased inflammation-induced vulnerability and decreased flexibility of the colonic tissue, the gut wall integrity might be more at risk during the postradiation recovery phase (i.e., until day 10) and upon the full manifestation of intestinal GvHD-related colitis (i.e., after approximately day 25). Importantly, however, we have not observed significantly increased complication rates dependent on the time point of endoscopy, as long as the endoscope is gently advanced under sufficient visibility. In contrast, we identified suboptimal drug dosing to be a risk factor since an insufficient depth of anesthesia can result in the occurrence of unwanted body movements of the experimental mouse and, consecutively, colonic wall perforation events. Second, the most critical step during the scoring process represents restrictions in this matter due to the unwanted presence of solid or liquid (e.g., diarrhea) feces in the gut lumen as the experimenter moves the endoscope into the scoring position. In this situation, flushing the colorectal region with saline solution using a flexible plastic pipette is often required. However, since this procedure might compromise the scoring accuracy and specificity of several parameters (e.g., stool consistency), this caveat carefully needs to be taken into consideration upon scoring.

There are several limitations restricting the interpretation and conclusions drawn from this endoscopic approach. First, intrinsic to a methodology using an inflexible endoscope via the rectal route, evaluations are limited to the last 3–4 cm of the gastrointestinal tract (i.e., the rectum and the distal colon). Therefore, GvHD-associated affections of the proximal colon and even the small intestine are by default excluded from the evaluation, indicating that scientific questions focusing on the small intestinal phenotype of GvHD will not benefit from this approach. Second, although inflammation is one of the hallmark morphologic features of intestinal GvHD, additional characteristics, like an increased apoptosis rate of the intestinal epithelial cells and a loss of the intestinal crypt architecture, are often found and included in the histopathological workflow underlying intestinal GvHD scoring and grading by pathologists in allo-HCT patients. Here, we restrictedly correlated the mini-endoscopically assessed intestinal GvHD scores with histopathologically assessed signs of inflammation. Taking this approach, we came to the conclusion that colonic inflammation above a certain threshold in the colonoscopic scoring is reliably predicting the presence of moderate-to-high grade inflammation, as assessed by histopathological postmortem analyses. However, future studies need to investigate how, for instance, histopathologically assessed apoptosis rates relate to the mini-endoscopic total sum scores or to, for instance, one of the four individual parameters embedded in the sum score. Third, while mice suffering from histopathologically moderate-to-high-grade GvHD can be readily identified by the endoscopic scoring approach, mice with more subtle signs of intestinal inflammation may be missed by the described approach, given the fact that histopathological scoring revealed the presence of mild signs of inflammation in the group of noT mice undergoing allo-HCT alone, without the transplantation of alloreactive donor lymphocytes. Biologically, this finding might be feasible, since noT mice have been allotransplanted, and minimal colitis may, indeed, reflect the presence of low levels of GvHD due to an incomplete removal of T cells from the bone marrow cell fraction or the reconstitution of alloreactive T cells from the bone marrow over time. Future studies need to address these questions in more detail. Fourth, formally, this protocol focuses on fully established intestinal GvHD-associated colitis. However, as shown and published previously, the onset of colitis around day 15 can be endoscopically detected and quantitated¹¹. Here, GvHD-prone mice could be clearly distinguished from noT mice in which the colonic phenotype might be solely due to residual signs of the irradiation-induced tissue damage or, rather, mirror the mucosal recovery response after irradiation¹¹. However, future studies need to correlate endoscopic evaluations and aim at detailed histopathological analyses of GvHD-prone mice compared to control mice at earlier time points.

Finally, an interesting, helpful add-on application of the endoscopic scoring of intestinal GvHD is the usage of the endoscopic working channel to direct a minimized biopsy forceps to the anatomic structure of choice in the colon. This feature provides, as a sort of built-in-option, the possibility to take view-guided tissue biopsies that can be further analyzed by a series of downstream techniques like histopathology, immunofluorescence staining, and real-time PCR analyses of genes of interest. However, future studies need to formally assess whether, for instance, histopathological scoring and grading of endoscopically taken biopsies are equivalent to histopathological results based on conventionally taken samples (i.e., from euthanized mice).

Overall, with this protocol describing the induction and its mini-endoscopic evaluation of intestinal GvHD, we provide a methodology to assess, score, and grade intestinal GvHD in a relevant murine GvHD model system of allo-HCT. The endoscopic assessment can be repeated in the same mouse at multiple time points over time, eliminating the need to euthanize mice to sequentially assess intestinal GvHD-associated tissue pathology (e.g., in a therapeutic setting). The mini-endoscopic scoring results proved to be comparable to results obtained by the histopathological assessment of intestinal inflammation in the colon and correlated well with systemic GvHD activity. Additionally, this procedure can be combined with view-guided biopsies via the working channel of the endoscope. Ultimately, however, future studies have to assess how far histopathologically determined morphological features other than inflammation, that are frequently found in intestinal GvHD-affected lesions, relate to the results obtained with this mini-endoscopic assessment and scoring approach.

Disclosures

The authors have nothing to disclose.

Acknowledgments

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Materials List for

Induction of Intestinal Graft-versus-host Disease and Its Mini-endoscopic Assessment in Live Mice

Vera Buchele^{1,3}, Maike Büttner-Herold², Tina Vogler^{1,3}, Markus F. Neurath^{1,3}, Kai Hildner^{1,3}

¹Department of Internal Medicine 1, University Erlangen-Nürnberg, University Hospital Erlangen ²Institute of Pathology, Department of Nephropathology, University Hospital Erlangen ³Deutsches Zentrum Immuntherapie (DZI), University Hospital Erlangen

Corresponding Author

Kai Hildner

Kai.Hildner@uk-erlangen.de

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Materials

Name	Company	Catalog Number	Comments
BIOBEAM 2000 Gamma Irradiator	Gamma-Service Meical GmbH		
Phosphate-buffered saline (PBS)	Sigma-Aldrich Co. LLC.	D8662-6x500ML	
Semken Forceps (length: 13 cm; serrated, curved tips)	Fine Science Tools	11009-13	
Hardened Fine Scissors (length: 8,5 cm; straight tips; cutting edge: 24 mm)	Fine Science Tools	14090-09	
RPMI-1640 Medium	Sigma-Aldrich Co. LLC.	R8758-500ML	
Hypodermic needle (26 G)	B. Braun Melsungen AG	4657683	
1 mL syring	B. Braun Melsungen AG	9166017V	
50 mL tube	Sarstedt Ag & Co.KG	62,547,254	
Cell strainer with a 40 µm mesh screen	BD Falcon	352340	
Ammonium chloride (NH ₄ Cl)	Sigma-Aldrich Co. LLC.	11209	ingredient of ACK lysing buffer
Ethylenediaminetetraacetic acid disodium salt dihydrate (Na ₂ EDTA)	Carl Roth GmbH & Co.KG	8043.2	ingredient of ACK lysing buffer
Potassium bicarbonate (KHCO ₃)	Merck KGaA	1,048,540,500	ingredient of ACK lysing buffer
CD90.2 MicroBeads, mouse	Miltenyi Biotec GmbH	130-049-101	magnet cell separation to isolate T cell-depleted bone marrow cells
Pan T Cell Isolation Kit II, mouse	Miltenyi Biotec GmbH	130-095-130	magnet cell separation to isolate splenic T cells
Alexa Fluor 700 anti-mouse CD45.2 Antibody (clone: 104; lot: B252126; RRID: AB_493731)	Biolegend	109822	
Pacific Blue anti-mouse CD3 Antibody (clone: 17A2; lot: B227246; RRID: AB_493645)	Biolegend	100214	
FITC anti-mouse CD4 Antibody (clone: GK1.5; lot: B225057; RRID: AB_312691)	Biolegend	100406	

PE/Cy7 anti-mouse CD45.1 Antibody (clone: A20; lot: B217246; RRID: AB_1134168)	Biolegend	110730	
APC/Cy7 anti-mouse CD8a Antibody (clone: 53-6.7, lot: B247008; RRID: AB_312753)	Biolegend	100714	
Filtrated bovine serum	Pan Biotec	P40-47500	ingredient of FACS buffer
96-well polystyrene V-bottom plates	Greiner Bio-One	651201	
Polystyrene Round-Bottom Tube (5 mL)	Falcon	352052	
BD LSRFortessa II flow cytometer	BD Bioscience Co.		
Insulin syringe with sterile interior (30 G)	BD	324826	
Oxy Vet Oxymat 3	Eickemeyer		oxygen concentrator for anesthesia
NarkoVet	Eickemeyer	213062	
Plexiglass chamber	Eickemeyer	214620	
Straight Forward Telescope	KARL STORZ SE &Co KG	64301 AA	part of the experimental setup for colonoscopy
Protection and Examination Sheath	KARL STORZ SE &Co KG	61029 C	part of the experimental setup for colonoscopy
Examination Sheath with working channel	KARL STORZ SE &Co KG	61029 D	part of the experimental setup for colonoscopy
Biopsy Forceps	KARL STORZ SE &Co KG	61071 ZJ	part of the experimental setup for colonoscopy
175 Watt SCB XenonLight Source	KARL STORZ SE &Co KG	20132120	part of the experimental setup for colonoscopy
Fiber Optic Light Cable	KARL STORZ SE &Co KG	495 NL	part of the experimental setup for colonoscopy
Image 1 S3 Camera Head	KARL STORZ SE &Co KG	22220030	part of the experimental setup for colonoscopy
Image 1 SCB Camera Control Unit	KARL STORZ SE &Co KG	22200020	part of the experimental setup for colonoscopy
LCD monitor	Olympus	OE181H	part of the experimental setup for colonoscopy
Forane / Isofluran	AbbVie Inc.	B506	
Formaldehyde solution 37 %	Carl Roth GmbH & Co.KG	7398.1	
5.0 mL Dispenser tip	Eppendorf AG	30089456	